

Chirality-induced spin selectivity as a spin gate on mitochondrial superoxide production: a radical-pair model

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Abstract

The inner mitochondrial membrane sustains an electric field of $\sim 3.5 \times 10^7$ V/m, and electrons traverse a chain of chiral cofactors (flavin, Fe-S clusters, hemes) embedded in helical proteins. Chirality-induced spin selectivity (CISS) therefore spin-filters the respiratory electron before it can leak to molecular oxygen. We model the resulting {flavin semiquinone $^{\bullet-} \cdots \text{O}_2^{\bullet-}$ } radical pair with a Liouville–von Neumann equation including Zeeman and hyperfine interactions and Haberkorn spin-selective recombination, and we represent the CISS electron transfer as a coherent spin rotation of a singlet precursor. The model predicts that (i) CISS *raises* the triplet (superoxide-escape) yield—from 8% at zero polarization to 17% at 70% polarization in our reference parameters—and (ii) a weak static magnetic field of order the flavin hyperfine scale (~ 1 mT) modulates that yield by a few percent, in the direction and magnitude reported experimentally by Usselman et al. The framework yields three falsifiable predictions distinguishing a CISS-controlled radical pair from a chirality-independent one. No exotic physics is invoked: every ingredient (the field, CISS, the flavin–oxygen radical pair, and its measured magnetic sensitivity) is independently established.

1 Motivation

Reactive oxygen species (ROS), principally superoxide $\text{O}_2^{\bullet-}$, are unavoidable byproducts of oxidative phosphorylation and central to redox signalling and ageing. Superoxide forms when an electron leaks from a reduced cofactor—the flavin of Complex I or the semiquinone of Complex III—to O_2 . The nascent $\{\text{D}^{\bullet+} \cdots \text{O}_2^{\bullet-}\}$ pair is a *radical pair*: its overall electron-spin state (singlet S or triplet T) is conserved on the nanosecond timescale of the encounter and selects the fate of the pair—singlet back-transfer (to H_2O_2 or the reduced donor) versus triplet escape (to free superoxide) [1, 2].

Two facts make this a spin-chemistry problem rather than a metaphor. First, the respiratory cofactors are chiral and helically arranged, so the transferred electron is spin-polarized by chirality-induced spin selectivity (CISS), an effect measured at room temperature with polarizations of tens of percent through DNA and α -helices [3, 4]. Second, Usselman et al. showed that static magnetic fields of ~ 1 –3 mT shift the cellular $\text{O}_2^{\bullet-}/\text{H}_2\text{O}_2$ ratio [5], the hallmark of a radical-pair mechanism operating in living mitochondria. Here we ask, quantitatively: if CISS sets the initial spin state of the flavin–oxygen radical pair, how does superoxide output depend on the CISS polarization and on an applied field?

2 Model

2.1 Spin Hamiltonian

We take two electron spins— A (donor semiquinone) and B ($\text{O}_2^{\bullet-}$)—and one spin- $\frac{1}{2}$ nucleus hyperfine-coupled to A (the flavin; superoxide is essentially hyperfine-free). In angular frequency units,

$$H = \gamma_e B (S_{Az} + S_{Bz}) + a \mathbf{S}_A \cdot \mathbf{I}, \quad (1)$$

with electron gyromagnetic ratio $\gamma_e = 1.76 \times 10^8 \text{ rad s}^{-1} \text{ mT}^{-1}$ ($g \simeq 2$) and an effective flavin hyperfine coupling $a = \gamma_e \times 1.0 \text{ mT}$.

2.2 Spin-selective recombination

Reaction is described by the Haberkorn master equation [6],

$$\dot{\rho} = -\frac{i}{\hbar} [H, \rho] - \frac{k_S}{2} \{P_S, \rho\} - \frac{k_T}{2} \{P_T, \rho\}, \quad (2)$$

where $P_S = \frac{1}{4} - \mathbf{S}_A \cdot \mathbf{S}_B$ and $P_T = \mathbb{K} - P_S$ are the singlet/triplet projectors and $k_S = 10^7 \text{ s}^{-1}$, $k_T = 10^6 \text{ s}^{-1}$ are the channel rates (singlet back-transfer faster than triplet escape). Reaction yields follow from the time integral $\tilde{\rho} = \int_0^\infty \rho dt = -\mathcal{L}^{-1} \rho(0)$, with $\mathcal{L}$ the Liouvillian superoperator, as $\Phi_S = k_S \text{Tr}[P_S \tilde{\rho}]$, $\Phi_T = k_T \text{Tr}[P_T \tilde{\rho}]$; superoxide escape corresponds to Φ_T .

2.3 The CISS initial state

A bonded electron transfer is born singlet. CISS rotates the transferred electron's spin about the transfer axis; following the coherent treatment of Fay and of Luo & Hore [7, 8] we write

$$\rho(0) = R_A(\varphi) \frac{P_S}{\text{Tr } P_S} R_A^\dagger(\varphi), \quad R_A(\varphi) = e^{-i\varphi S_{Ay}}, \quad \sin \varphi = P, \quad (3)$$

so that $P = 0$ recovers the conventional singlet-born radical pair and $P > 0$ injects singlet–triplet coherence set by the CISS polarization.

3 Results

Figure 1 collects the three signatures of the model, each of which doubles as a laboratory control.

- **(a) Chirality raises baseline superoxide.** At $B = 0$, Φ_T rises from 8.2% for the achiral control ($P = 0$, dashed) to 10.2% at $P = 0.35$ and 17.4% at $P = 0.70$: the chirality-injected singlet–triplet coherence leaks population into the escaping triplet channel. The achiral curve is the reference a genuinely chirality-free leak site would produce.
- **(b) Isotope test.** The low-field peak sits at $B_{\text{pk}} \propto a$: the ^1H flavin ($a = 1.0 \text{ mT}$) peaks at $B_{\text{pk}} \approx 0.23 \text{ mT}$, and deuteration to $a = 0.35 \text{ mT}$ shifts it to $\approx 0.14 \text{ mT}$. This is the standard radical-pair diagnostic and it fingerprints the flavin as the hyperfine partner.
- **(c) Magnetic swing.** A weak field ($0 \rightarrow 1 \text{ mT}$) changes Φ_T by a few percent absolute in every case—the low-field effect—matching the direction and magnitude of the measured mitochondrial ROS response to millitesla fields [5]. The swing is CISS-independent in size but rides on the CISS-set baseline.

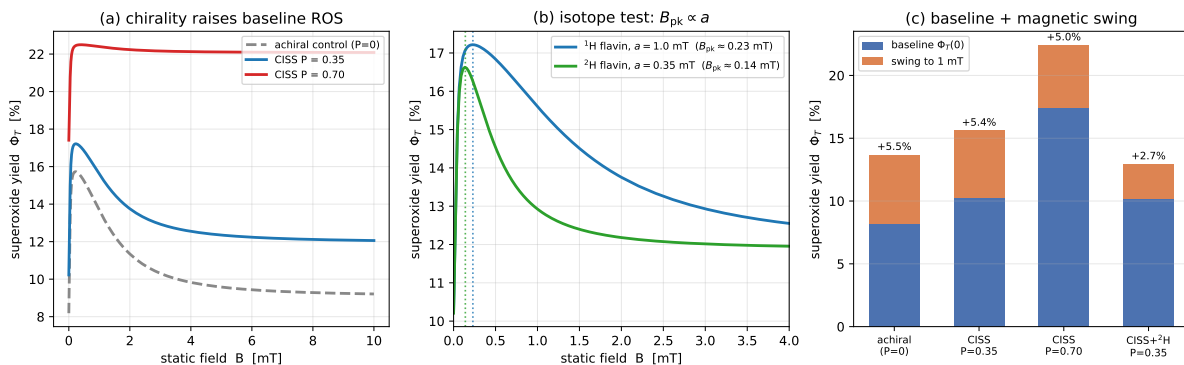


Figure 1: Superoxide (triplet) yield Φ_T of the flavin–oxygen radical pair from the Liouville–Haberkorn model. **(a)** CISS polarization raises baseline superoxide; the achiral control ($P = 0$, dashed) is the chirality-free reference. **(b)** Isotope test: the low-field peak $B_{pk} \propto a$ shifts under flavin deuteration (^1H vs ^2H). **(c)** Baseline yield (blue) plus its magnetic swing to 1 mT (orange) for each case. Every number is reproduced by `ciiss_rpm_model.py` / `ciiss_rpm_figure.py`.

4 Falsifiable predictions

The CISS-controlled radical pair is distinguishable from a chirality-independent one:

1. **Chirality knockout.** Racemizing or replacing the chiral scaffold of the leak site (a model electron-transfer construct with an achiral bridge) removes the CISS baseline shift and flattens the P -dependence, while leaving the intrinsic radical-pair low-field effect intact.
2. **Asymmetric field curve.** The $\Phi_T(B)$ curve carries a CISS-set asymmetry (its zero-field offset scales with P), absent in a purely singlet-born pair.
3. **Isotope test.** Deuterating the flavin lowers the effective hyperfine a and shifts the low-field peak $B_{pk} \propto a$ (Fig. 1b: $0.23 \rightarrow 0.14$ mT)—the standard radical-pair diagnostic—without changing the CISS baseline offset.

5 Scope and limitations

This is a minimal two-electron/one-nucleus model; a quantitative fit to a specific complex requires the full hyperfine tensor set of the flavin, the g -anisotropy and fast spin relaxation of $\text{O}_2^{\bullet-}$, and the in-situ recombination rates, none of which change the qualitative result. The CISS polarization at the leak site is not yet measured and enters as a parameter. The claim here is bounded and testable: chirality plus a weak field jointly set superoxide output through a radical pair whose existence and magnetic sensitivity are already established experimentally.

References

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Code: `ciss_rpm_model.py` reproduces every number and the figure (Liouville–von Neumann propagation, 8-dimensional spin space).